

Project Topic Selection – Krisztina Fabo - Spring 2026

DBMS for GRC Risk and Compliance Tracking System

Description

Organizations today must manage increasing levels of operational, regulatory, and security risk. As regulatory requirements grow more complex and the volume of internal controls increases, traditional spreadsheet-based or flat-file systems become insufficient for tracking risks, monitoring control effectiveness, and ensuring compliance. These limitations make it difficult to maintain data consistency, quickly retrieve information, and generate meaningful reports for decision-making.

The goal of this project is to design and implement a relational database management system (DBMS) that supports Governance, Risk, and Compliance (GRC) activities within an organization. The system will provide a structured and centralized repository for managing risks, controls, compliance requirements, audit activities, and responsible personnel. By organizing this information within a relational schema and providing a web-based interface, the system will enable efficient data entry, retrieval, and analysis.

The proposed system will allow users to record risks along with their severity and status, associate risks with mitigating controls, and link controls to compliance requirements derived from regulatory frameworks. In addition, the system will track audit activities used to evaluate control effectiveness and store information about employees or departments responsible for managing risks and controls. The system will support queries and reports that help identify high-risk areas, unmet compliance requirements, and audit findings, thereby improving visibility and supporting informed decision-making.

Scope

The scope of this project is limited to the design and implementation of a core GRC tracking system using a relational database and a web-based user interface. The database will capture structured information about the following components:

- Risks, including their descriptions, severity levels, and current status
- Controls, including their type, implementation status, and ownership
- Compliance requirements, including framework references and requirement details
- Audit records, including audit dates and evaluation results
- Employees and/or departments responsible for managing risks and controls

The system will model the relationships among these components, such as the association between risks and controls, controls and compliance requirements, and audits and evaluated controls. These relationships will be implemented using appropriate relational schemas and constraints.

The web interface will provide functionality for inserting new data (e.g., adding risks or controls), displaying data through queries that combine multiple relations, and searching for records based on specified criteria (e.g., risk level or control status). The interface will be designed to be simple, navigable, and sufficient to demonstrate interaction with the underlying database.

This project will not include advanced features such as real-time monitoring, automated risk scoring, user authentication systems, or integration with external data sources.